

Sufficient Statistics for Economic Mobility: When Do Measures Agree?

Rory McGee
UWO & IFS

May 2026
UM-MSU-UWO Labo(u)r Day

A “proliferation of measures”

- Researchers measure economic mobility
 - ... in many outcomes (e.g., incomes, wealth, education, health, ...)
 - ... for many uses (e.g., normative analysis, policy evaluation, validating models, ...)
 - ... for many comparisons (e.g., cross-country, regions, history, ...)

A “proliferation of measures”

- Researchers measure economic mobility
 - ... in many outcomes (e.g., incomes, wealth, education, health, ...)
 - ... for many uses (e.g., normative analysis, policy evaluation, validating models, ...)
 - ... for many comparisons (e.g., cross-country, regions, history, ...)
- ...**and in many ways**: intergenerational elasticity (IGE), rank-rank slopes, transition matrices, directional mobility, axiomatic indices (e.g., Shorrocks)

A “proliferation of measures”

- Researchers measure economic mobility
 - ... in many outcomes (e.g., incomes, wealth, education, health, ...)
 - ... for many uses (e.g., normative analysis, policy evaluation, validating models, ...)
 - ... for many comparisons (e.g., cross-country, regions, history, ...)
- ...**and in many ways**: intergenerational elasticity (IGE), rank-rank slopes, transition matrices, directional mobility, axiomatic indices (e.g., Shorrocks)
- Different measures often produce consistent rankings of mobility (e.g., Katz and Krueger 2017, Berman 2022, Deutscher and Mazumder 2023)

A “proliferation of measures”

- Researchers measure economic mobility
 - ... in many outcomes (e.g., incomes, wealth, education, health, ...)
 - ... for many uses (e.g., normative analysis, policy evaluation, validating models, ...)
 - ... for many comparisons (e.g., cross-country, regions, history, ...)
- ...and in many ways: intergenerational elasticity (IGE), rank-rank slopes, transition matrices, directional mobility, axiomatic indices (e.g., Shorrocks)
- Different measures often produce consistent rankings of mobility (e.g., Katz and Krueger 2017, Berman 2022, Deutscher and Mazumder 2023)
- But not always: long-run US trends (Jácome, Kuziemko & Naidu, 2025); cross-CZ w/in US (Chetty et al., 2014); cross-country (Blanden 2013) ▶ CZs

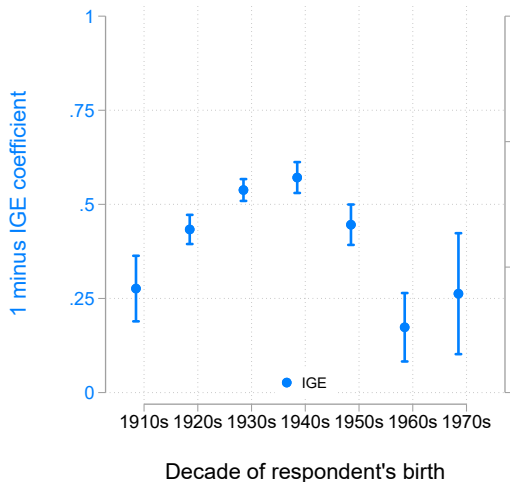
A Puzzle: 20th Century U.S. Intergenerational Mobility

Did it **Rise** or **Fall**? (Jácome, Kuziemko & Naidu, 2025)

A Puzzle: 20th Century U.S. Intergenerational Mobility

Did it **Rise** or **Fall**? (Jácome, Kuziemko & Naidu, 2025)

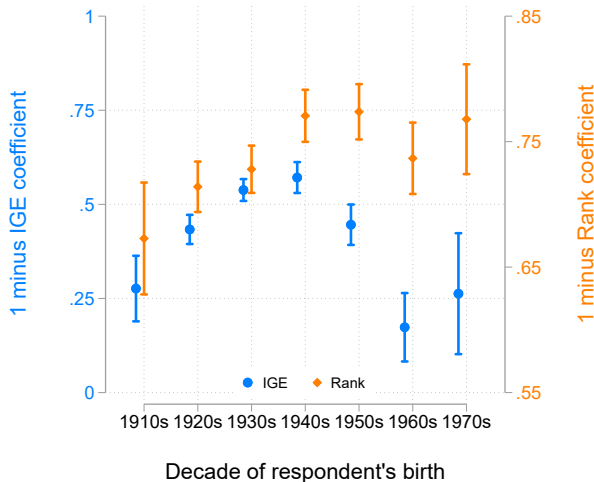
- **IGE**: rose $\uparrow$ after 1950
→ Mobility *fell* $\downarrow$



A Puzzle: 20th Century U.S. Intergenerational Mobility

Did it **Rise** or **Fall**? (Jácome, Kuziemko & Naidu, 2025)

- **IGE**: rose $\uparrow$ after 1950
→ Mobility *fell* $\downarrow$
- **Rank-rank slope**: fell $\downarrow$ after 1950
→ Mobility *rose* $\uparrow$



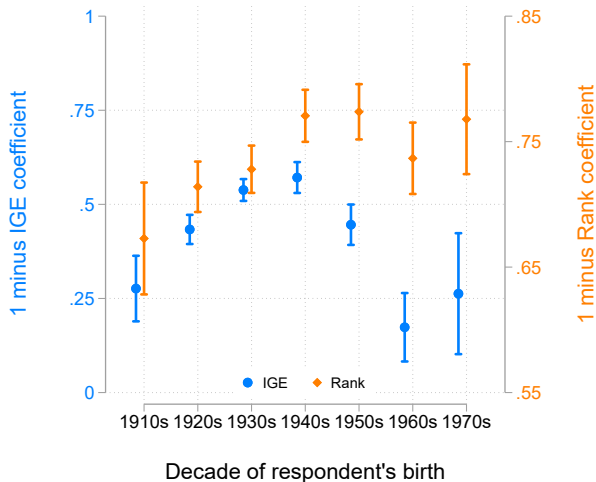
A Puzzle: 20th Century U.S. Intergenerational Mobility

Did it **Rise** or **Fall**? (Jácome, Kuziemko & Naidu, 2025)

- **IGE**: rose $\uparrow$ after 1950
→ Mobility *fell* $\downarrow$
- **Rank-rank slope**: fell $\downarrow$ after 1950
→ Mobility *rose* $\uparrow$

Two fundamental questions:

1. When *should* we expect measures to agree?
2. What does disagreement *tell us*?



This Paper Provides Answers

1. Agreement of measures depends on single property of parent-child joint distribution
 - **Concordance order:** parent-child ranks are *everywhere* more aligned

This Paper Provides Answers

1. Agreement of measures depends on single property of parent-child joint distribution
 - **Concordance order**: parent-child ranks are *everywhere* more aligned
2. If economies ranked by concordance, **all standard mobility measures agree**
 - Observe ranking in one measure, know all rankings: measures as ordinal substitutes
 - Fixing marginal distributions of income

This Paper Provides Answers

1. Agreement of measures depends on single property of parent-child joint distribution
 - **Concordance order**: parent-child ranks are *everywhere* more aligned
2. If economies ranked by concordance, **all standard mobility measures agree**
 - Observe ranking in one measure, know all rankings: measures as ordinal substitutes
 - Fixing marginal distributions of income
3. Changes in any mobility measure = rank-dependence + changing inequality
 - Rank-based measures: only rank-dependence
 - Level-based (incl. IGE): both $\longrightarrow$ potential for disagreement
 - Disagreement reflects:
 - i. different weightings over regions in joint distribution
 - ii. different sensitivity to changing inequality

Contributions

1. Unifying framework w/ **Concordance** as sufficient statistic for mobility measurement
 - Adds to measure comparisons (Deutscher, Mazumder; Berman), axiomatic approaches (Shorrocks; Fields, Ok; D'Agostino, Dardanoni; Cowell, Flachaire), dual of 2D-inequality problem (Atkinson, Bourguignon; Dardanoni)
 - Scope of debate over preferred measures
 - Complement inference (Chetverikov, Wilhelm; Kitagawa, Nybom, Stuhler; Haider, Solon) w/ results for proxies and measurement error

Contributions

1. Unifying framework w/ **Concordance** as sufficient statistic for mobility measurement
 - Adds to measure comparisons (Deutscher, Mazumder; Berman), axiomatic approaches (Shorrocks; Fields, Ok; D'Agostino, Dardanoni; Cowell, Flachaire), dual of 2D-inequality problem (Atkinson, Bourguignon; Dardanoni)
 - Scope of debate over preferred measures
 - Complement inference (Chetverikov, Wilhelm; Kitagawa, Nybom, Stuhler; Haider, Solon) w/ results for proxies and measurement error
2. Empirical validation & new decomposition explaining measure disagreement
 - Clarifies IGE vs rank-rank divergence (Jácome, Kuziemko, Naidu; Blanden; Corak)
 - Diagnostic tool for applied researchers

Contributions

1. Unifying framework w/ **Concordance** as sufficient statistic for mobility measurement
 - Adds to measure comparisons (Deutscher, Mazumder; Berman), axiomatic approaches (Shorrocks; Fields, Ok; D'Agostino, Dardanoni; Cowell, Flachaire), dual of 2D-inequality problem (Atkinson, Bourguignon; Dardanoni)
 - Scope of debate over preferred measures
 - Complement inference (Chetverikov, Wilhelm; Kitagawa, Nybom, Stuhler; Haider, Solon) w/ results for proxies and measurement error
2. Empirical validation & new decomposition explaining measure disagreement
 - Clarifies IGE vs rank-rank divergence (Jácome, Kuziemko, Naidu; Blanden; Corak)
 - Diagnostic tool for applied researchers
3. Microfounded w/ structural interpretation in Becker-Tomes-Loury models
 - Sorting measures NAM vs PAM in matching (Anderson, Smith; Gola; Boerma, Tsyvinski, Wang, Zhang) and mating (Chiappori, Costa Dias, Meghir, Zhang)

Setup

Separating Dependence from Marginals: Copulas

- Child and parent incomes (Y^K, Y^P) have joint CDF $F(Y^K, Y^P)$

Separating Dependence from Marginals: Copulas

- Child and parent incomes (Y^K, Y^P) have joint CDF $F(Y^K, Y^P)$
 - Uniform ranks $R^K = F_K(Y^K)$, $R^P = F_P(Y^P)$ on $[0, 1]$

Separating Dependence from Marginals: Copulas

- Child and parent incomes (Y^K, Y^P) have joint CDF $F(Y^K, Y^P)$
 - Uniform ranks $R^K = F_K(Y^K)$, $R^P = F_P(Y^P)$ on $[0, 1]$

Sklar's (1959) Theorem: Any joint distribution decomposes as:

$$F(Y^K, Y^P) = C(F_K(Y^K), F_P(Y^P))$$

where $C : [0, 1]^2 \rightarrow [0, 1]$ is the **copula** or joint distribution of ranks

Separating Dependence from Marginals: Copulas

- Child and parent incomes (Y^K, Y^P) have joint CDF $F(Y^K, Y^P)$
 - Uniform ranks $R^K = F_K(Y^K)$, $R^P = F_P(Y^P)$ on $[0, 1]$

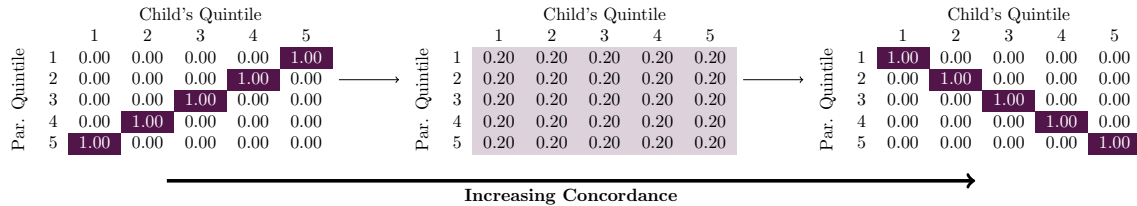
Sklar's (1959) Theorem: Any joint distribution decomposes as:

$$F(Y^K, Y^P) = C(F_K(Y^K), F_P(Y^P))$$

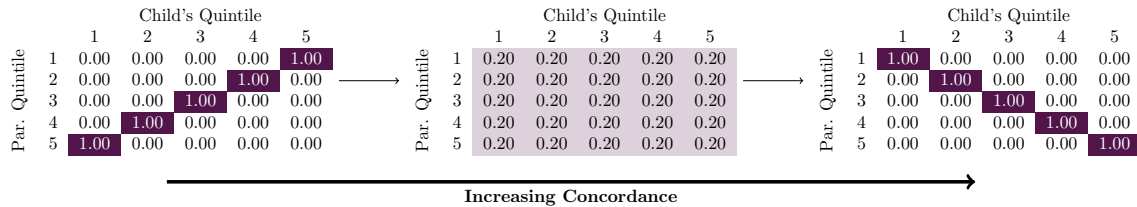
where $C : [0, 1]^2 \rightarrow [0, 1]$ is the **copula** or joint distribution of ranks

- Copula encodes *all* rank-dependence between generations $\perp$ of marginals ► Mass points
- Relative mobility measures alternative summaries of copula

Concordance Order: Visual Intuition ▶ Example II

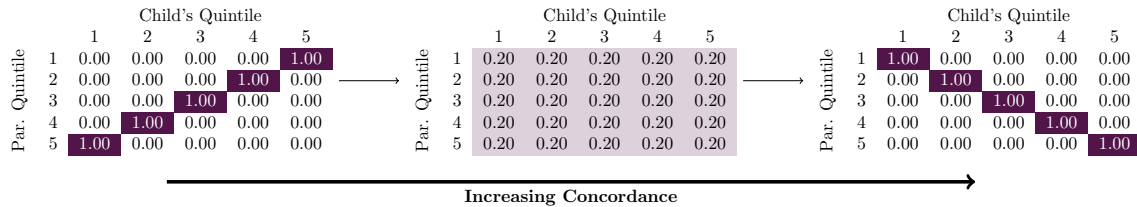


Concordance Order: Visual Intuition ▶ Example II



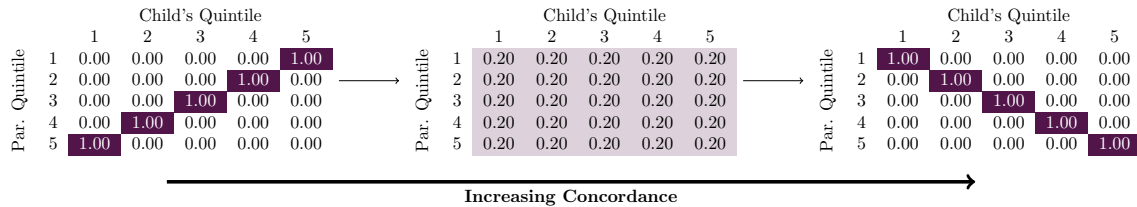
- Shift mass from off-diagonal (rank-reversing) to diagonal (rank-preserving) cells

Concordance Order: Visual Intuition ▶ Example II



- Shift mass from off-diagonal (rank-reversing) to diagonal (rank-preserving) cells
- “Large” parent values coincide with “large” child values more often

Concordance Order: Visual Intuition ▶ Example II



- Shift mass from off-diagonal (rank-reversing) to diagonal (rank-preserving) cells
- “Large” parent values coincide with “large” child values more often
- Pinned down by dependence parameters in parametric families (e.g., ρ in normal)

The Concordance Order

Definition: (Tchen 1980) Economy A is **more concordant** than economy B ($C^A \succeq C^B$) if:

$$C^A(u, v) \geq C^B(u, v) \quad \forall u, v \in [0, 1]$$

The Concordance Order

Definition: (Tchen 1980) Economy A is **more concordant** than economy B ($C^A \succeq C^B$) if:

$$C^A(u, v) \geq C^B(u, v) \quad \forall u, v \in [0, 1]$$

- Higher concordance = parent-child ranks **more aligned everywhere**
 - A form of stochastic dominance for *dependence*

The Concordance Order

Definition: (Tchen 1980) Economy A is **more concordant** than economy B ($C^A \succeq C^B$) if:

$$C^A(u, v) \geq C^B(u, v) \quad \forall u, v \in [0, 1]$$

- Higher concordance = parent-child ranks **more aligned everywhere**
 - A form of stochastic dominance for *dependence*
 - Incomplete order over distributions, but covers empirically relevant cases [▶ jump e.g.](#)
 - “intuitive requirement that children from higher-income families are less likely to have lower incomes” (Chetty et al., 2017); “empirically unlikely transition matrices” (Shorrocks, 1978); “all-or-nothing” (Berman, 2022)
 - Results extend to approximate concordance (continuous)

The Concordance Order

Definition: (Tchen 1980) Economy A is **more concordant** than economy B ($C^A \succeq C^B$) if:

$$C^A(u, v) \geq C^B(u, v) \quad \forall u, v \in [0, 1]$$

- Higher concordance = parent-child ranks **more aligned everywhere**
 - A form of stochastic dominance for *dependence*
 - Incomplete order over distributions, but covers empirically relevant cases [▶ jump e.g.](#)
 - “intuitive requirement that children from higher-income families are less likely to have lower incomes” (Chetty et al., 2017); “empirically unlikely transition matrices” (Shorrocks, 1978); “all-or-nothing” (Berman, 2022)
 - Results extend to approximate concordance (continuous)
- Testable: provide transparent testing procedure

The Concordance Order

Definition: (Tchen 1980) Economy A is **more concordant** than economy B ($C^A \succeq C^B$) if:

$$C^A(u, v) \geq C^B(u, v) \quad \forall u, v \in [0, 1]$$

- Higher concordance = parent-child ranks **more aligned everywhere**
 - A form of stochastic dominance for *dependence*
 - Incomplete order over distributions, but covers empirically relevant cases [▶ jump e.g.](#)
 - “intuitive requirement that children from higher-income families are less likely to have lower incomes” (Chetty et al., 2017); “empirically unlikely transition matrices” (Shorrocks, 1978); “all-or-nothing” (Berman, 2022)
 - Results extend to approximate concordance (continuous)
 - Testable: provide transparent testing procedure
- Higher concordance $\longleftrightarrow$ **lower mobility**

Main Results: Sufficiency

Concordance Orders All Standard Mobility Measures

Props 1-4: Fix marginals. If $C^A \succeq C^B$, then economy A is *less mobile* than B by:

1. Regression/Correlation-based measures:
 - IGE: $\beta^A \geq \beta^B$, Rank-rank slope: $\rho^A \geq \rho^B$, Intergenerational correlation
2. “Local” measures:
 - Conditional expected rank $\longrightarrow$ rotates around $r^* = 0.5$
3. Transition probabilities:
 - diagonal $\uparrow$, off-diagonal $\downarrow$, Directional mobility
4. Axiomatic measures:
 - Shorrocks trace, Fields & Ok, D’Agostino & Dardanoni, Cowell & Flachaire

Concordance Orders All Standard Mobility Measures

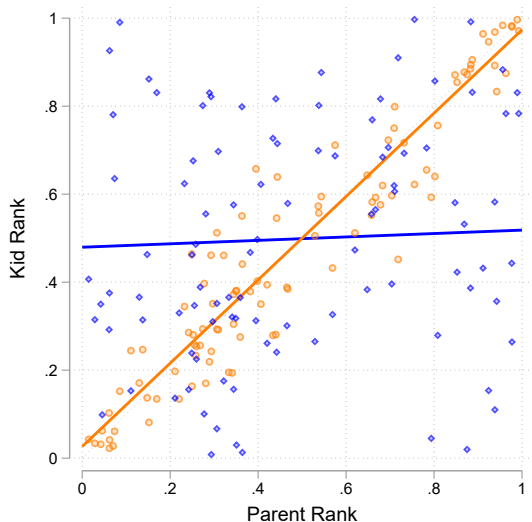
Props 1-4: Fix marginals. If $C^A \succeq C^B$, then economy A is *less mobile* than B by:

1. Regression/Correlation-based measures:
 - IGE: $\beta^A \geq \beta^B$, Rank-rank slope: $\rho^A \geq \rho^B$, Intergenerational correlation
2. “Local” measures:
 - Conditional expected rank $\rightarrow$ *rotates* around $r^* = 0.5$
3. Transition probabilities:
 - diagonal $\uparrow$, off-diagonal $\downarrow$, Directional mobility
4. Axiomatic measures:
 - Shorrocks trace, Fields & Ok, D’Agostino & Dardanoni, Cowell & Flachaire

$\rightarrow$ Measures are **ordinal substitutes**: observe one ranking, know all others

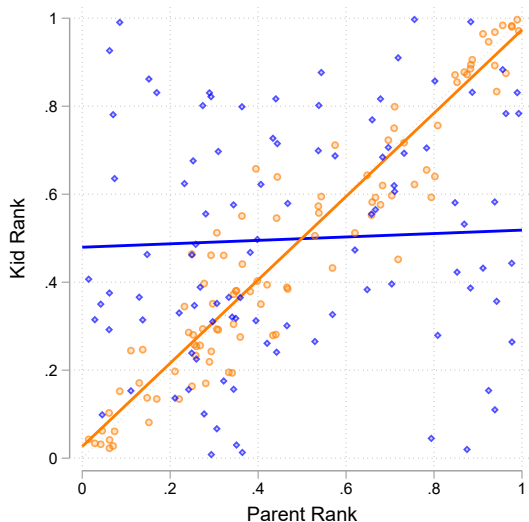
Why Does Concordance Order These Measures?

Rank-rank slope



Why Does Concordance Order These Measures?

Rank-rank slope



Mobility Measure:

$$\rho = 12 \iint C(u, v) du dv - 3$$

$C^A \geq C^B$ everywhere

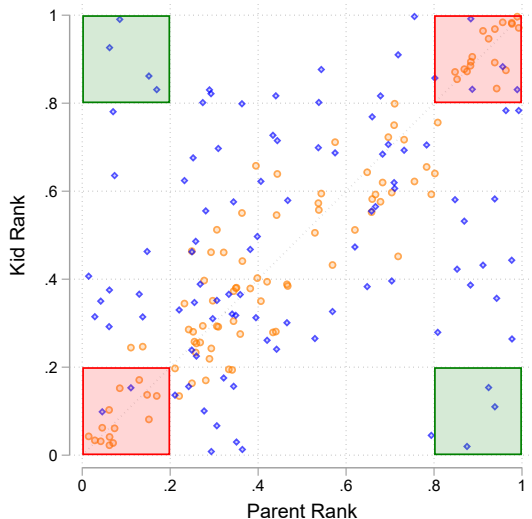
$$\longrightarrow \iint C^A \geq \iint C^B$$

$$\longrightarrow \rho^A \geq \rho^B$$

(Lehmann 1966; Tchen 1980)

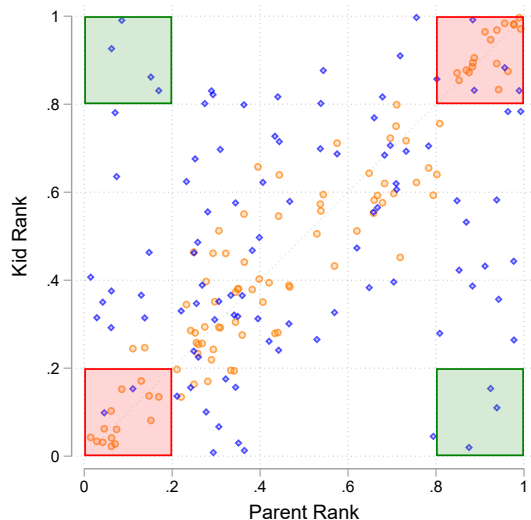
Why Does Concordance Order These Measures?

Transition Probabilities (on-diagonal)



Why Does Concordance Order These Measures?

Transition Probabilities (on-diagonal)



Mobility Measure:

$$\Pr(R^K > \tau \mid R^P > \tau) = \frac{1 - 2\tau + C(\tau, \tau)}{1 - \tau}$$

If $C^A \geq C^B$ everywhere

$$\longrightarrow C^A(\tau, \tau) \geq C^B(\tau, \tau)$$

$$\longrightarrow \text{TP}^A \geq \text{TP}^B$$

(Copula definition, simplest cases!)



Concordance is Pigou-Dalton for Mobility

- Inequality measurement (Atkinson, 1970)
 1. Elementary operation: Pigou-Dalton transfer
 2. Partial order: Lorenz dominance (SOSD)
 3. Sufficiency: SOSD $\longrightarrow$ all transfer-sensitive measures agree

Concordance is Pigou-Dalton for Mobility

- Inequality measurement (Atkinson, 1970)
 1. Elementary operation: Pigou-Dalton transfer
 2. Partial order: Lorenz dominance (SOSD)
 3. Sufficiency: SOSD $\rightarrow$ all transfer-sensitive measures agree
- Mobility measurement (Today)
 1. Elementary operation: Concordance Exchanges
 - swap Y_i^K and Y_j^K if $Y_i^K < Y_j^K$, but $Y_i^P > Y_j^P \rightarrow$ increase concordance
 2. Partial order: Concordance
 - Discuss implications & completions
 3. Sufficiency: Concordance $\rightarrow$ all exchange mobility measures agree

Analogues in SWF for 2D ineq. (Atkinson-Bourguignon) & Perm. Income ineq. (Dardanoni)

U.S. Mobility Trends & Formal Test

Empirical Application: 20th Century U.S. Mobility

Datasets: Replicate **Jácome, Kuziemko & Naidu, 2025**

- Pool 15 U.S. surveys (ANES, GSS, PSID, OCG, NLS, ...) with respondent income + father's occupation (large sample of intergenerational links) 
- Impute parental income from Census using occupation $\times$ race $\times$ region (using census)
- U.S. Born, age 30-50 (mitigate life-cycle bias), 1910s-1970s birth cohort (YoB)
- 56,102 father-son pairs

Empirical Application: 20th Century U.S. Mobility

Datasets: Replicate *Jácome, Kuziemko & Naidu, 2025*

- Pool 15 U.S. surveys (ANES, GSS, PSID, OCG, NLS, ...) with respondent income + father's occupation (large sample of intergenerational links) ▶
- Impute parental income from Census using occupation $\times$ race $\times$ region (using census)
- U.S. Born, age 30-50 (mitigate life-cycle bias), 1910s-1970s birth cohort (YoB)
- 56,102 father-son pairs

Imputation **does not** affect theory insights ▶

- *bias levels* but *ordering preserved*

Empirical Application: 20th Century U.S. Mobility

Datasets: Replicate *Jácome, Kuziemko & Naidu, 2025*

- Pool 15 U.S. surveys (ANES, GSS, PSID, OCG, NLS, ...) with respondent income + father's occupation (large sample of intergenerational links) ▶
- Impute parental income from Census using occupation $\times$ race $\times$ region (using census)
- U.S. Born, age 30-50 (mitigate life-cycle bias), 1910s-1970s birth cohort (YoB)
- 56,102 father-son pairs

Imputation **does not** affect theory insights ▶

- *bias levels* but *ordering preserved*

Theory predicts:

- Rank-based measures should **co-move**, but other measures may diverge (rising inequality)

Visualizing Approximate Concordance

Construct cardinal measure M_λ for (approx) concordance order

Visualizing Approximate Concordance

Construct cardinal measure M_λ for (approx) concordance order

- Mix perfect immobility (Hoeffding bound M) and perfect mobility ($C^{\mathbb{I}}$)

$$M_\lambda = \lambda^* \in \operatorname{argmin}_{\lambda \in [0,1]} \int_0^1 \int_0^1 \left[C^{obs}(u, v) - \left((1 - \lambda) M(u, v) + \lambda C_{uv}^{\mathbb{I}} \right) \right]^2 du dv$$

- λ^* is weight on origin-independence (Prais, 1955): 0 to 1

Visualizing Approximate Concordance

Construct cardinal measure M_λ for (approx) concordance order

- Mix perfect immobility (Hoeffding bound M) and perfect mobility ($C^{\mathbb{I}}$)

$$M_\lambda = \lambda^* \in \operatorname{argmin}_{\lambda \in [0,1]} \int_0^1 \int_0^1 \left[C^{obs}(u, v) - \left((1 - \lambda) M(u, v) + \lambda C_{uv}^{\mathbb{I}} \right) \right]^2 du dv$$

- λ^* is weight on origin-independence (Prais, 1955): 0 to 1
- Parallels PAM approach in marriage market (Fernández and Rogerson, 2001; Abbott et al., 2019)

Visualizing Approximate Concordance

Construct cardinal measure M_λ for (approx) concordance order

- Mix perfect immobility (Hoeffding bound M) and perfect mobility ($C^{\mathbb{I}}$)

$$M_\lambda = \lambda^* \in \operatorname{argmin}_{\lambda \in [0,1]} \int_0^1 \int_0^1 \left[C^{obs}(u, v) - \left((1 - \lambda) M(u, v) + \lambda C_{uv}^{\mathbb{I}} \right) \right]^2 du dv$$

- λ^* is weight on origin-independence (Prais, 1955): 0 to 1
- Parallels PAM approach in marriage market (Fernández and Rogerson, 2001; Abbott et al., 2019)
- Robust to violations

Visualizing Approximate Concordance

Construct cardinal measure M_λ for (approx) concordance order

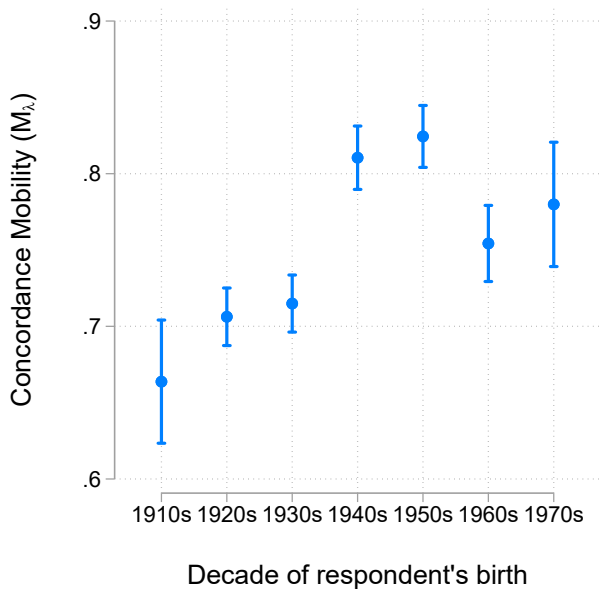
- Mix perfect immobility (Hoeffding bound M) and perfect mobility ($C^{\mathbb{I}}$)

$$M_\lambda = \lambda^* \in \operatorname{argmin}_{\lambda \in [0,1]} \int_0^1 \int_0^1 \left[C^{obs}(u, v) - \left((1 - \lambda) M(u, v) + \lambda C_{uv}^{\mathbb{I}} \right) \right]^2 du dv$$

- λ^* is weight on origin-independence (Prais, 1955): 0 to 1
- Parallels PAM approach in marriage market (Fernández and Rogerson, 2001; Abbott et al., 2019)
- Robust to violations

Contrast with results from formal testing procedure

Concordance Mobility over the 20th Century



Wave-pattern

- Long run Mobility rises over the 20th century
- But plateaus/declines after 1960s

Increasing Mobility? Moment Inequality Test of Concordance Order

► Details

- Intuition $\approx$ Kolmogorov-Smirnov test for Stochastic Dominance

Increasing Mobility? Moment Inequality Test of Concordance Order

► Details

- Intuition $\approx$ Kolmogorov-Smirnov test for Stochastic Dominance
- Long Run test vs 1910? **Increased Mobility!**

Increasing Mobility? Moment Inequality Test of Concordance Order

► Details

- Intuition $\approx$ Kolmogorov-Smirnov test for Stochastic Dominance
- Long Run test vs 1910? **Increased Mobility!**
- Comparing successive decades?

Increasing Mobility? Moment Inequality Test of Concordance Order

► Details

- Intuition $\approx$ Kolmogorov-Smirnov test for Stochastic Dominance
- Long Run test vs 1910? **Increased Mobility!**
- Comparing successive decades?

Decade A	Decade B	$pval_{A \succeq B}$	$pval_{B \succeq A}$	Interpretation
1910	1920	.231	.034	$C_{1910} \succeq C_{1920}$
1920	1930	.001	.004	Copulas cross
1930	1940	.052	.001	$C_{1930} \succeq C_{1940}$
1940	1950	.142	.358	Fail to reject
1950	1960	.004	.045	Copulas cross
1960	1970	.190	.017	$C_{1960} \succeq C_{1970}$

Increasing Mobility? Moment Inequality Test of Concordance Order

► Details

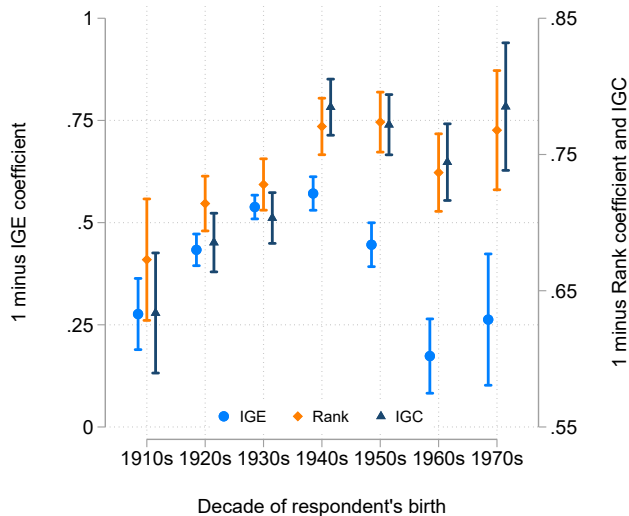
- Intuition $\approx$ Kolmogorov-Smirnov test for Stochastic Dominance
- Long Run test vs 1910? **Increased Mobility!**
- Comparing successive decades?

Decade A	Decade B	$pval_{A \succeq B}$	$pval_{B \succeq A}$	Interpretation
1910	1920	.231	.034	$C_{1910} \succeq C_{1920}$
1920	1930	.001	.004	Copulas cross
1930	1940	.052	.001	$C_{1930} \succeq C_{1940}$
1940	1950	.142	.358	Fail to reject
1950	1960	.004	.045	Copulas cross
1960	1970	.190	.017	$C_{1960} \succeq C_{1970}$

- Some measure disagreement (paper has many measures) noise, others substantive
→ Transition Probabilities disagree when copulas cross

Regression Measures Don't Comove

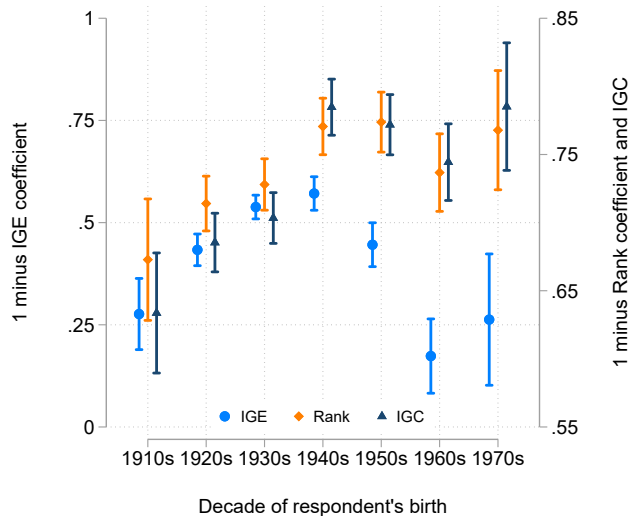
► Other Measures



- Rank-rank slope inherits wave-like pattern from concordance

Regression Measures Don't Comove

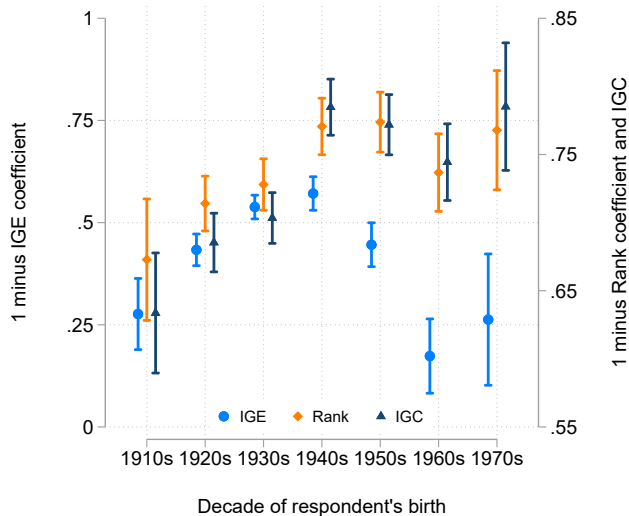
► Other Measures



- Rank-rank slope inherits wave-like pattern from concordance
- IGC tracks rank-rank slope (not generic)

Regression Measures Don't Comove

► Other Measures



- Rank-rank slope inherits wave-like pattern from concordance
- IGC tracks rank-rank slope (not generic)
- IGE diverges starting 1950s

Why? IGE adds changing inequality to theory results

Extension: When Marginals Differ ▶ e.g.

- Comparisons across time/space typically involve different marginal distributions

Extension: When Marginals Differ ▶ e.g.

- Comparisons across time/space typically involve different marginal distributions
- Any measure $M = M(C, F_K, F_P)$ can be decomposed:

$$\Delta M = M^B - M^A = \underbrace{M(C^B, F_K^B, F_P^B) - M(C^A, F_K^B, F_P^B)}_{\text{Copula component (varying dependence)}} + \underbrace{M(C^A, F_K^B, F_P^B) - M(C^A, F_K^A, F_P^A)}_{\text{Marginal component (varying marginals)}}$$

Extension: When Marginals Differ ▶ e.g.

- Comparisons across time/space typically involve different marginal distributions
- Any measure $M = M(C, F_K, F_P)$ can be decomposed:

$$\Delta M = M^B - M^A = \underbrace{M(\mathbf{C}^B, \mathbf{F}_K^B, \mathbf{F}_P^B) - M(\mathbf{C}^A, \mathbf{F}_K^B, \mathbf{F}_P^B)}_{\text{Copula component (varying dependence)}}$$

$$+ \underbrace{M(C^A, F_K^B, F_P^B) - M(C^A, F_K^A, F_P^A)}_{\text{Marginal component (varying marginals)}}$$

Extension: When Marginals Differ ▶ e.g.

- Comparisons across time/space typically involve different marginal distributions
- Any measure $M = M(C, F_K, F_P)$ can be decomposed:

$$\begin{aligned}\Delta M = M^B - M^A &= \underbrace{M(C^B, F_K^B, F_P^B) - M(C^A, F_K^B, F_P^B)}_{\text{Copula component (varying dependence)}} \\ &+ \underbrace{M(C^A, F_K^B, F_P^B) - M(C^A, F_K^A, F_P^A)}_{\text{Marginal component (varying marginals)}}\end{aligned}$$

Extension: When Marginals Differ ▶ e.g.

- Comparisons across time/space typically involve different marginal distributions
- Any measure $M = M(C, F_K, F_P)$ can be decomposed:

$$\Delta M = M^B - M^A = \underbrace{M(C^B, F_K^B, F_P^B) - M(C^A, F_K^B, F_P^B)}_{\text{Copula component (varying dependence)}} + \underbrace{M(C^A, F_K^B, F_P^B) - M(C^A, F_K^A, F_P^A)}_{\text{Marginal component (varying marginals)}}$$

- Rank-based measures: Marginal component = 0 by construction!

Extension: When Marginals Differ ▶ e.g.

- Comparisons across time/space typically involve different marginal distributions
- Any measure $M = M(C, F_K, F_P)$ can be decomposed:

$$\begin{aligned}\Delta M = M^B - M^A = & \underbrace{M(C^B, F_K^B, F_P^B) - M(C^A, F_K^B, F_P^B)}_{\text{Copula component (varying dependence)}} \\ & + \underbrace{M(C^A, F_K^B, F_P^B) - M(C^A, F_K^A, F_P^A)}_{\text{Marginal component (varying marginals)}}\end{aligned}$$

- Rank-based measures: Marginal component = 0 by construction!
- Level-based measures (IGE): Both components non-zero & potentially opposites

Extension: When Marginals Differ ▶ e.g.

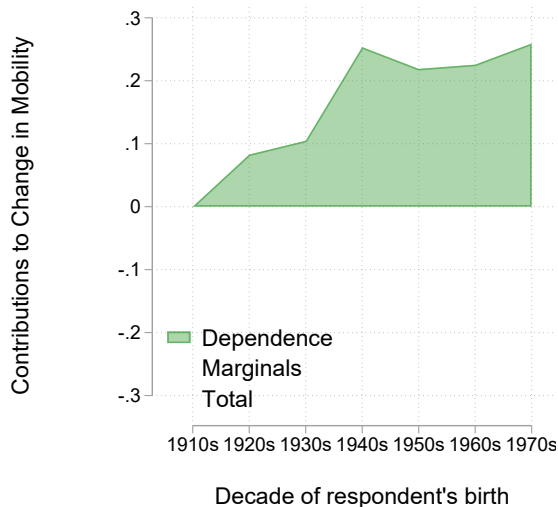
- Comparisons across time/space typically involve different marginal distributions
- Any measure $M = M(C, F_K, F_P)$ can be decomposed:

$$\begin{aligned}\Delta M = M^B - M^A = & \underbrace{M(C^B, F_K^B, F_P^B) - M(C^A, F_K^B, F_P^B)}_{\text{Copula component (varying dependence)}} \\ & + \underbrace{M(C^A, F_K^B, F_P^B) - M(C^A, F_K^A, F_P^A)}_{\text{Marginal component (varying marginals)}}\end{aligned}$$

- Rank-based measures: Marginal component = 0 by construction!
- Level-based measures (IGE): Both components non-zero & potentially opposites
- Order matters $\longrightarrow$ average over fixing copulas at A and B
(Shapley-Owen-Shorrocks decomposition)

Decomposition: Why Does IGE Diverge?

► Other Measures



- **Dependence component:** rising mobility + plateau
 - Consistent with M_λ and R-R slope
 - Those are only dependence

Decomposition: Why Does IGE Diverge?

► Other Measures

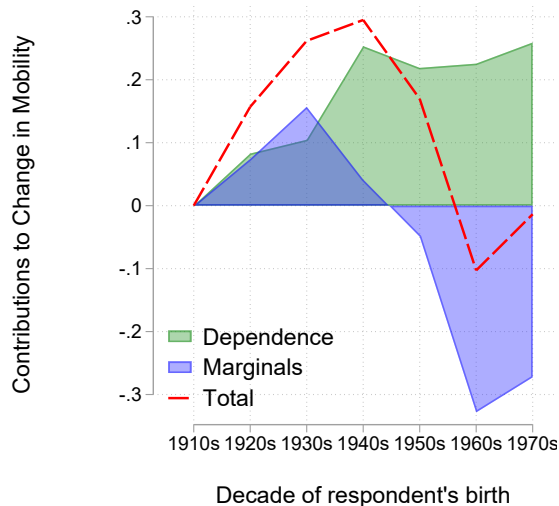


- **Dependence component:** rising mobility + plateau
 - Consistent with M_λ and R-R slope
 - Those are only dependence
- **Marginal component:** rising inequality pulls mobility down

$$-\frac{\sigma_K^{1960}/\sigma_P^{1960}}{\sigma_K^{1910}/\sigma_P^{1910}} \approx 1.75$$

Decomposition: Why Does IGE Diverge?

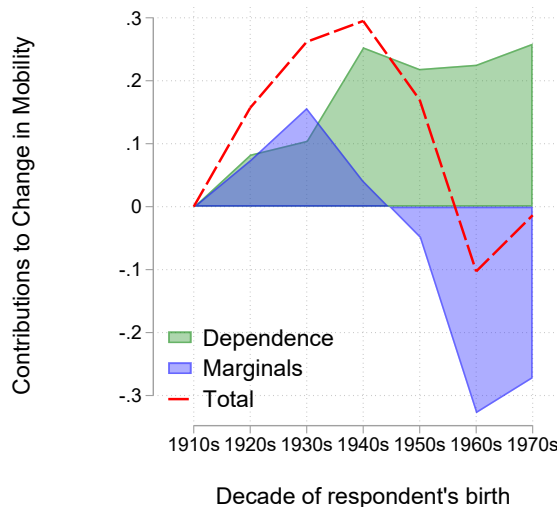
► Other Measures



- **Dependence component:** rising mobility + plateau
 - Consistent with M_λ and R-R slope
 - Those are only dependence
- **Marginal component:** rising inequality pulls mobility down
 - $\frac{\sigma_K^{1960}/\sigma_P^{1960}}{\sigma_K^{1910}/\sigma_P^{1910}} \approx 1.75$
- Attenuates post-1940 & dominates by 1970

Decomposition: Why Does IGE Diverge?

► Other Measures



- **Dependence component:** rising mobility + plateau
 - Consistent with M_λ and R-R slope
 - Those are only dependence
- **Marginal component:** rising inequality pulls mobility down
 - $$\frac{\sigma_K^{1960}/\sigma_P^{1960}}{\sigma_K^{1910}/\sigma_P^{1910}} \approx 1.75$$
- Attenuates post-1940 & dominates by 1970
- Disagreement reflects sensitivity to marginals!

What To Do When Concordance Fails?

Crossing Copulas: 1920s–30s and 1950s–60s ◀▶

Concordance is a partial order → not all economies are ranked, e.g.

		Child's Quintile				
		1	2	3	4	5
Par. Quintile	1	0.32	0.24	0.2	0.15	0.08
	2	0.25	0.23	0.21	0.18	0.12
	3	0.19	0.22	0.22	0.21	0.16
	4	0.15	0.18	0.21	0.24	0.22
	5	0.09	0.12	0.16	0.22	0.41

No
Concordance
Order
←→

		Child's Quintile				
		1	2	3	4	5
Par. Quintile	1	0.58	0.00	0.00	0.14	0.28
	2	0.00	0.86	0.00	0.00	0.14
	3	0.00	0.00	1.00	0.00	0.00
	4	0.14	0.00	0.00	0.86	0.00
	5	0.28	0.14	0.00	0.00	0.58

- Identical rank-correlation (0.32)

Source of Disagreement

- Measures expressed as

$$M = \int \int \phi_M(Y^K, Y^P) f(Y^K, Y^P) dY^K dY^P,$$

where $\phi_M(Y^K, Y^P)$: mobility-measure specific intensity

- TP: Indicator; IGE: $1/\sigma_P^2(\ln Y^K - \ln \bar{Y}^K)(\ln Y^P - \ln \bar{Y}^P)$

Source of Disagreement

- Measures expressed as

$$M = \int \int \phi_M(Y^K, Y^P) f(Y^K, Y^P) dY^K dY^P,$$

where $\phi_M(Y^K, Y^P)$: mobility-measure specific intensity

- TP: Indicator; IGE: $1/\sigma_P^2(\ln Y^K - \ln \bar{Y}^K)(\ln Y^P - \ln \bar{Y}^P)$
- For fixed marginals:
$$\Delta M = \int_0^1 \int_0^1 \underbrace{W_M(R^K, R^P; F^K, F^P)}_{\leq 0} \left[C^A(R^K, R^P) - C^B(R^K, R^P) \right] dR^K dR^P$$
- Concordance guarantees $C^A - C^B \geq 0 \longrightarrow \Delta M \geq 0 \perp$ of weights 

Source of Disagreement

- Measures expressed as

$$M = \int \int \phi_M(Y^K, Y^P) f(Y^K, Y^P) dY^K dY^P,$$

where $\phi_M(Y^K, Y^P)$: mobility-measure specific intensity

- TP: Indicator; IGE: $1/\sigma_P^2(\ln Y^K - \ln \bar{Y}^K)(\ln Y^P - \ln \bar{Y}^P)$
- For fixed marginals:
$$\Delta M = \int_0^1 \int_0^1 \underbrace{W_M(R^K, R^P; F^K, F^P)}_{\leq 0} \left[C^A(R^K, R^P) - C^B(R^K, R^P) \right] dR^K dR^P$$
- Concordance guarantees $C^A - C^B \geq 0 \longrightarrow \Delta M \geq 0 \perp$ of weights 
- Failures $\longrightarrow C^A - C^B$ changes sign on diff. regions, but measures diff. sensitivity
 - Tie-break: **Normative** (intersecting Lorenz curves) or **robustness** choice (sensitivity to ϵ -violations)

What To Report When Concordance Fails

Prop 6: Concordance isn't knife-edge!

If $C^A - C^B$ within ϵ of satisfying concordance a measure M agrees up to $L_M \cdot \epsilon$

- L_M small for rank-based measures, larger for IGE and specific TPs
- Rationalises tighter empirical co-movement of rank-based measures

What To Report When Concordance Fails

Prop 6: Concordance isn't knife-edge!

If $C^A - C^B$ within ϵ of satisfying concordance a measure M agrees up to $L_M \cdot \epsilon$

- L_M small for rank-based measures, larger for IGE and specific TPs
 - Rationalises tighter empirical co-movement of rank-based measures

Reporting protocol: use test output to localise the violation

- Report **local rank-rank slopes** on the regions where $C^A - C^B$ changes sign
- Supplement with transition probabilities at standard thresholds

Lets readers with different preferred measures reach their own conclusions

Other Extensions & Structural Interpretation

No time? Jump to Conclusions

Other Settings ▶ Skip

1. More measures (follow JEL taxonomy, Deutscher & Mazumder) include absolute measures
 - Lorenz mobility (McGee & Ocampo, 2025), Bartholomew jump measure, Goodman, average non-linear income persistence (Arellano, Blundell, Bonhomme, 2017), Katz & Krueger
2. Concordance order maintained under fixed rank comparisons
 - Ranking children by position in parent distributions
 - Subgroup or subnational regions against population distribution
3. Allow for multi-dimensional mobility
 - Whole lifecycle paths (Audoly, McGee, Ocampo & Paz-Pardo, 2025) or many generations

Microfoundation: Becker-Tomes-Loury

- Standard dynastic investment model

$$\max_{C^P, I} U(C^P) + \delta \mathbb{E}_\theta [U(C^K)]$$

subject to

$$C^P = Y^P - I, \quad C^K = Y^K = W \left(\underbrace{\theta f(I, H^P)}_{H^K} \right)$$

- Parents choose investment (I) and consumption (C^P); child consumes (C^K)
- $f(\cdot)$: skill production technology; increasing and (weakly) supermodular
- θ : luck shock realized after investment

Microfoundation: Becker-Tomes-Loury

- Standard dynastic investment model

$$\max_{C^P, I} U(C^P) + \delta \mathbb{E}_\theta [U(C^K)]$$

subject to

$$C^P = Y^P - I, \quad C^K = Y^K = W \left(\underbrace{\theta f(I, H^P)}_{H^K} \right)$$

- Parents choose investment (I) and consumption (C^P); child consumes (C^K)
- $f(\cdot)$: skill production technology; increasing and (weakly) supermodular
- θ : luck shock realized after investment
- When do changes in technology or preferences lead to more mobility?

Investment Expansions $\longrightarrow$ Larger in Concordance ▶ Example

Lemma 1: If investment expands uniformly ▶

$$I^{*A}(Y^P) \geq I^{*B}(Y^P) \quad \forall Y^P \longrightarrow C^A \succeq C^B$$

Investment Expansions $\longrightarrow$ Larger in Concordance ▶ Example

Lemma 1: If investment expands uniformly ▶

$$I^{*A}(Y^P) \geq I^{*B}(Y^P) \quad \forall Y^P \longrightarrow C^A \succeq C^B$$

Mechanisms (Corollary 3) ▶

1. Higher marginal product of investment, $f_I^A(I, H^P) \geq f_I^B(I, H^P) \quad \forall (H^P, I)$ or subsidy;
2. Higher returns to human capital, $W^A(H^K) \geq W^B(H^K) \quad \forall H^K$;
3. Higher discount factor, $\delta^A \geq \delta^B$.

Investment Expansions → Larger in Concordance ▶ Example

Lemma 1: If investment expands uniformly ▶

$$I^{*A}(Y^P) \geq I^{*B}(Y^P) \quad \forall Y^P \longrightarrow C^A \succeq C^B$$

Mechanisms (Corollary 3) ▶

1. Higher marginal product of investment, $f_I^A(I, H^P) \geq f_I^B(I, H^P) \quad \forall (H^P, I)$ or subsidy;
2. Higher returns to human capital, $W^A(H^K) \geq W^B(H^K) \quad \forall H^K$;
3. Higher discount factor, $\delta^A \geq \delta^B$.

What about heterogeneous policy impacts?

- E.g. school finance equalisation: poor districts $\uparrow$, rich districts $\downarrow$ (Biasi, 2023)
- $\Delta I^*(Y^P)$ crosses zero $\longrightarrow \Delta C$ changes sign $\longrightarrow$ crossing copulas
- Rationalises 1920s–30s, 1950s–60s?

Conclusions

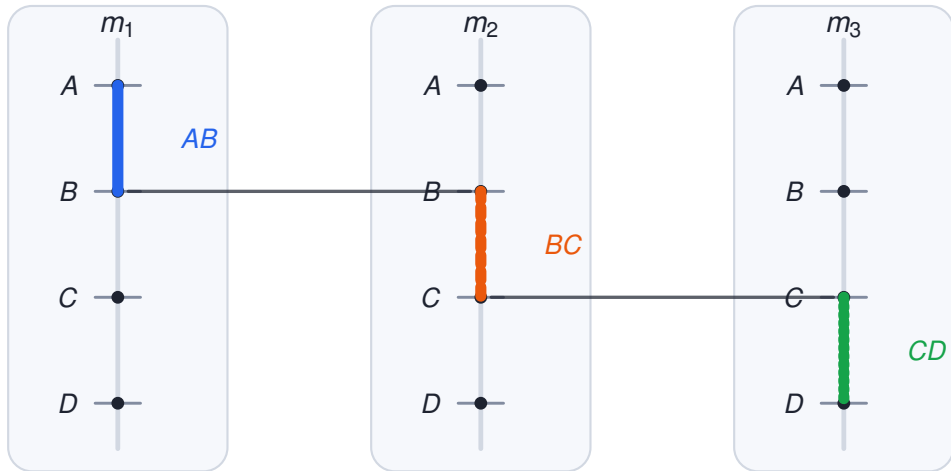
Implications For Applied Researchers

A single property—**concordance**—determines when standard mobility measures agree

For applied researchers:

1. **When concordance holds:** measures are substitutes
 - Pick by data quality, not axiom: transitions for noisy data, IGE for clean panels, rank-rank for cross-country
2. **When it fails:** disagreement is informative, not noise
 - Test it; report local rank-rank slopes; Lipschitz bounds for ϵ -violations
3. **Decomposition as diagnostic:** rank-dependence vs. changing inequality
(reframes IGE-vs-rank-rank divergence)

Chaining Measures



Gives full comparison of A, B, C and D in *any* measure

Details

e.g. from bunching, top-coding, censoring, genuine discreteness $\longrightarrow$ no restrictions on F_K or F_P , but Copula 'indeterminate'

Solutions:

1. Parametric copulas

- Does violence to discrete data
- Restrictive

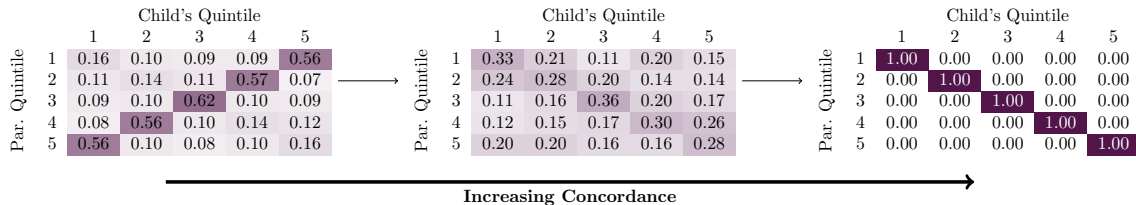
2. Carley (2002) bounds on copulas

- A lot more math, but flexible
- Worst-case bounds may not be tight

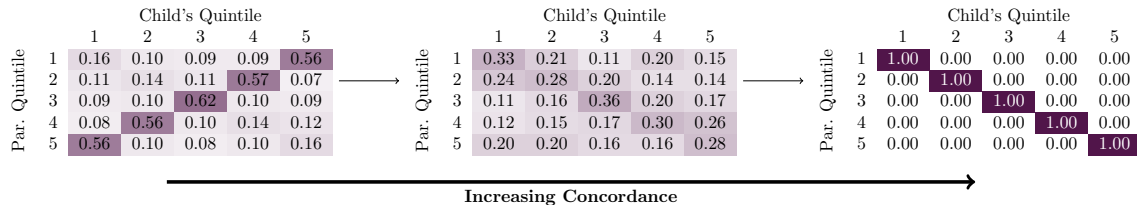
3. Pick an extension: checkerboard or Piecewise bilinear

- Propositions 11, 13 in Genest and Nešlehová (2007) show the unique extension preserving concordance order of the original joint-distributions

Concordance: Visual Intuition

[◀ Back](#)

Concordance: Visual Intuition [◀ Back](#)



- Not just extreme cases

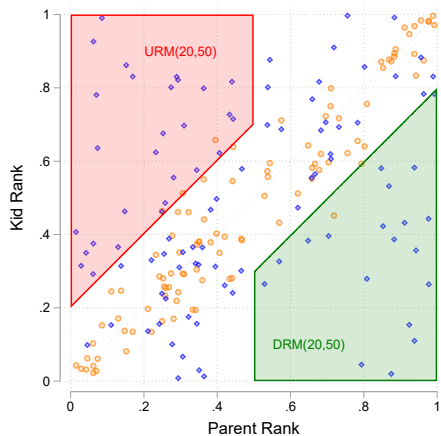
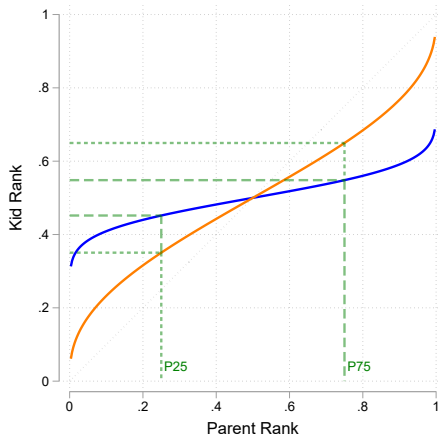
Some results require extra structure

1. Single crossing of conditional densities and continuity (sufficient: TP2 or MLRP)
 - For Non-parametric CER rotation
2. Smooth Anti-Diagonal to Diagonal Exchanges:

$$\Delta C(u_1, v_1) + \Delta C(u_2, v_2) \geq \Delta C(u_2, v_1) + \Delta C(u_1, v_2).$$

- Implies Concordance Order
- Directional Mobility & Shorrocks

Why Does Concordance Order These Measures? Illustration III

[Back](#)

Extension: Proxies & Measurement Error [◀ Back](#)

Prop 5: Observe proxy $E^g = H_g(R^g, \varepsilon^g)$ for generation g , H_g increasing, and ε independent of ranks. Let $\tilde{C}$ denote copula for (E^K, E^P) . **Concordance ordering preserved:**

$$C^A \succeq C^B \quad \longrightarrow \quad \tilde{C}^A \succeq \tilde{C}^B$$

Extension: Proxies & Measurement Error [◀ Back](#)

Prop 5: Observe proxy $E^g = H_g(R^g, \varepsilon^g)$ for generation g , H_g increasing, and ε independent of ranks. Let $\tilde{C}$ denote copula for (E^K, E^P) . **Concordance ordering preserved:**

$$C^A \succeq C^B \quad \longrightarrow \quad \tilde{C}^A \succeq \tilde{C}^B$$

→ Error may **bias levels** but **ordering preserved**

Extension: Proxies & Measurement Error [◀ Back](#)

Prop 5: Observe proxy $E^g = H_g(R^g, \varepsilon^g)$ for generation g , H_g increasing, and ε independent of ranks. Let $\tilde{C}$ denote copula for (E^K, E^P) . **Concordance ordering preserved:**

$$C^A \succeq C^B \quad \longrightarrow \quad \tilde{C}^A \succeq \tilde{C}^B$$

→ Error may **bias levels** but **ordering preserved**

- Covers common proxy cases 

1. Education/occupation as income proxies: $Y \longrightarrow E$
2. Income and Education as skill proxies: $S \longrightarrow E, Y$

Extension: Proxies & Measurement Error

Prop 5: Observe proxy $E^g = H_g(R^g, \varepsilon^g)$ for generation g , H_g increasing, and ε independent of ranks. Let $\tilde{C}$ denote copula for (E^K, E^P) . **Concordance ordering preserved:**

$$C^A \succeq C^B \quad \longrightarrow \quad \tilde{C}^A \succeq \tilde{C}^B$$

→ Error may **bias levels** but **ordering preserved**

- Covers common proxy cases 

1. Education/occupation as income proxies: $Y \longrightarrow E$

2. Income and Education as skill proxies: $S \longrightarrow E, Y$

- Covers flexible measurement error cases $\tilde{Y} = E = H_g(R^g, \varepsilon^g)$ (Corollary 1) 

1. (non-)Classical measurement error in (log-)incomes (Bound, Brown, Duncan & Rogers 1994)

2. Measurement error in ranks (Chetverikov & Wilhelm 2023)

3. Life cycle biases (Kitagawa, Nybom, & Stuhler 2018)

Proxies & Meas Error: Functional Forms [< Back](#)

Proxies:

1. Human capital determines incomes $Y = H(E) + \varepsilon$ with $E = H^{-1}(Y - \varepsilon)$
2. Ability determines education and income $E = H_E(S, \varepsilon)$ and $Y = H_Y(S, \varepsilon)$.

Proxies & Meas Error: Functional Forms [◀ Back](#)

Proxies:

1. Human capital determines incomes $Y = H(E) + \varepsilon$ with $E = H^{-1}(Y - \varepsilon)$
2. Ability determines education and income $E = H_E(S, \varepsilon)$ and $Y = H_Y(S, \varepsilon)$.

Measurement error:

1. Classical measurement error in incomes

$$\tilde{Y}^K = F_K^{-1}(R^K) + \varepsilon^K, \quad \tilde{Y}^P = F_P^{-1}(R^P) + \varepsilon^P$$

2. Measurement error in ranks

$$\tilde{R}^K = F_{\tilde{K}} \left(F_K^{-1}(R^K) + \varepsilon^K \right), \quad \tilde{R}^P = F_{\tilde{P}} \left(F_P^{-1}(R^P) + \varepsilon^P \right)$$

3. Life Cycle Biases

$$\tilde{Y}^K = G_K \left(F_K^{-1}(R^K) + \mu_{\varepsilon^K} + \sigma_{\varepsilon^K} \varepsilon^K \right), \quad \tilde{Y}^P = G_P \left(F_P^{-1}(R^P) + \mu_{\varepsilon^P} + \sigma_{\varepsilon^P} \varepsilon^P \right)$$

Measure specific weights [◀ Back](#)

- Measures expressed as

$$M_M = \int \int \phi_M(Y^K, Y^P) f(Y^K, Y^P) dY^K dY^P,$$

where $\phi_M(Y^K, Y^P)$: mobility-measure specific intensity

- For fixed marginals, any twice-differentiable measure: $M_M^A - M_M^B$

$$= \int_0^1 \int_0^1 W_M(R^K, R^P; F^K, F^P) [C^A(R^K, R^P) - C^B(R^K, R^P)] dR^K dR^P$$

where

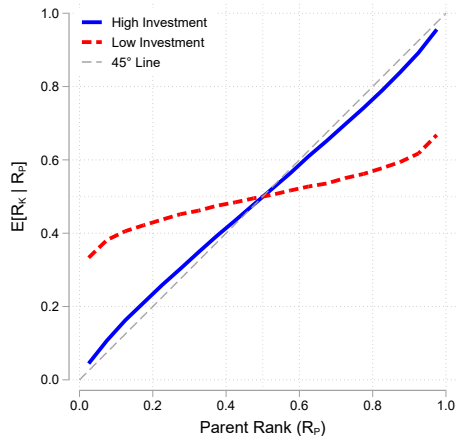
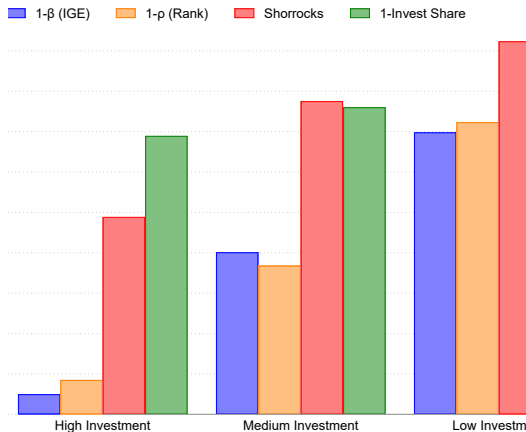
$$W_M(R^K, R^P) = \frac{\phi_{M,12}(F_K^{-1}(R^K), F_P^{-1}(R^P))}{f_K(F_K^{-1}(R^K)) f_P(F_P^{-1}(R^P))}$$

- Works for non-continuous cases also

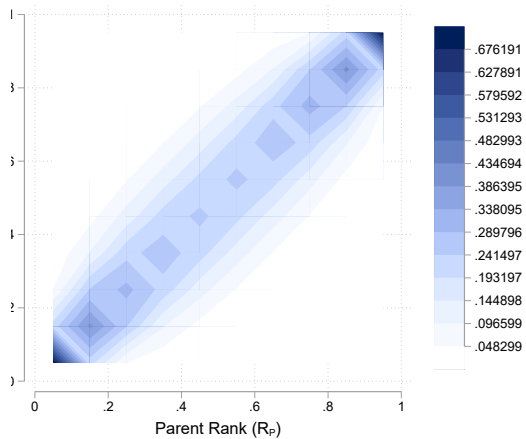
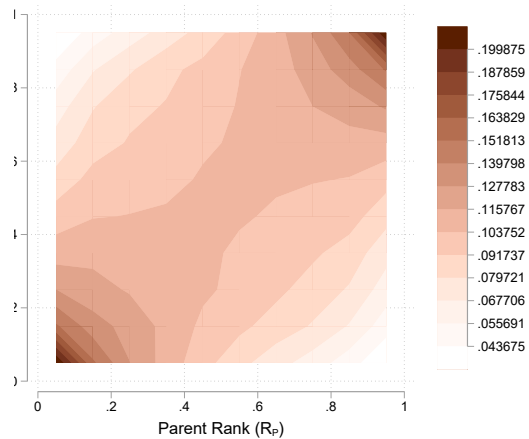
Investment Expansions Example

[◀ Back](#)

Log-utility and Cobb-Douglas Production



Investment Expansions Example

[◀ Back](#)

Proof Sketch: Lemma (Investment $\longrightarrow$ Concordance) [◀ Back](#)

- $I^{*A}(Y^P) \geq I^{*B}(Y^P)$ for all Y^P

Proof Sketch: Lemma (Investment $\longrightarrow$ Concordance) [◀ Back](#)

- $I^{*A}(Y^P) \geq I^{*B}(Y^P)$ for all Y^P
- f is increasing in l :

$$H^{K,A} \mid Y^P = y \succeq_{\text{FOSD}} H^{K,B} \mid Y^P = y \quad \forall y$$

Proof Sketch: Lemma (Investment $\longrightarrow$ Concordance) [◀ Back](#)

- $I^{*A}(Y^P) \geq I^{*B}(Y^P)$ for all Y^P
- f is increasing in I :

$$H^{K,A} \mid Y^P = y \succeq_{\text{FOSD}} H^{K,B} \mid Y^P = y \quad \forall y$$

- Under common CDF, rank variables satisfy:

$$F_{R^{K,A} \mid Y^P}(r \mid y) \leq F_{R^{K,B} \mid Y^P}(r \mid y) \quad \forall r \in (0, 1), \forall y$$

Proof Sketch: Lemma (Investment $\longrightarrow$ Concordance) [< Back](#)

- $I^{*A}(Y^P) \geq I^{*B}(Y^P)$ for all Y^P
- f is increasing in I :

$$H^{K,A} \mid Y^P = y \succeq_{\text{FOSD}} H^{K,B} \mid Y^P = y \quad \forall y$$

- Under common CDF, rank variables satisfy:

$$F_{R^{K,A} \mid Y^P}(r \mid y) \leq F_{R^{K,B} \mid Y^P}(r \mid y) \quad \forall r \in (0, 1), \forall y$$

- But CDF not common, 3 cases:
 1. *Rank-dependent wages*: Integrate over $Y^P \Rightarrow (R^K, Y^P)^A \succeq (R^K, Y^P)^B$
 2. *Common wage function*: FOSD of child incomes + decomposition theorem $\longrightarrow$ concordance $\uparrow$
 3. *Increased returns*: Analogous reasoning applies

Proof Sketch: Corollary (Policy $\longrightarrow$ Investment) [◀ Back](#)

Define value function for investment I and 'policy' λ :

$$V(I; \lambda) = U(C^P) + \delta_\lambda \mathbb{E}_\theta[U(W_\lambda(H^K))]$$

subsume subsidy or transfer into f

Proof Sketch: Corollary (Policy $\longrightarrow$ Investment) [◀ Back](#)

Define value function for investment I and 'policy' λ :

$$V(I; \lambda) = U(C^P) + \delta_\lambda \mathbb{E}_\theta [U(W_\lambda(H^K))]$$

subsume subsidy or transfer into f

Key condition (Topkis): Increasing differences holds if

$$\frac{\partial^2 V}{\partial I \partial \lambda} = \frac{\partial}{\partial \lambda} \left[\delta_\lambda \frac{\partial f_\lambda}{\partial I} \mathbb{E}_\theta [U'(W_\lambda) W'_\lambda] \right] \geq 0$$

Proof Sketch: Corollary (Policy $\longrightarrow$ Investment) [< Back](#)

Define value function for investment I and 'policy' λ :

$$V(I; \lambda) = U(C^P) + \delta_\lambda \mathbb{E}_\theta [U(W_\lambda(H^K))]$$

subsume subsidy or transfer into f

Key condition (Topkis): Increasing differences holds if

$$\frac{\partial^2 V}{\partial I \partial \lambda} = \frac{\partial}{\partial \lambda} \left[\delta_\lambda \frac{\partial f_\lambda}{\partial I} \mathbb{E}_\theta [U'(W_\lambda) W'_\lambda] \right] \geq 0$$

Verification:

- Production function f is supermodular
- Wage schedule W_λ is strictly increasing
- Standard regularity on GE effects (e.g., multiplicative shifts)

Proof Sketch: Corollary (Policy $\longrightarrow$ Investment) [< Back](#)

Define value function for investment I and 'policy' λ :

$$V(I; \lambda) = U(C^P) + \delta_\lambda \mathbb{E}_\theta [U(W_\lambda(H^K))]$$

subsume subsidy or transfer into f

Key condition (Topkis): Increasing differences holds if

$$\frac{\partial^2 V}{\partial I \partial \lambda} = \frac{\partial}{\partial \lambda} \left[\delta_\lambda \frac{\partial f_\lambda}{\partial I} \mathbb{E}_\theta [U'(W_\lambda) W'_\lambda] \right] \geq 0$$

Verification:

- Production function f is supermodular
- Wage schedule W_λ is strictly increasing
- Standard regularity on GE effects (e.g., multiplicative shifts)

Conclusion: $\lambda' < \lambda \implies I^*(Y^P; \lambda') \leq I^*(Y^P; \lambda)$

Jácome, Kuziemko & Naidu (2025) Replication [◀ Back](#)

15 Pooled Surveys:

- General Social Survey (GSS), American National Election Survey (ANES)
- Panel Study of Income Dynamics (PSID), Occupational Changes in a Generation (OCG)
- National Longitudinal Surveys (NLS): Mature Women, Older Men, Younger Women
- National Survey of Black Americans, National Fertility Survey, others

Sample Construction:

- Inclusion: US-born, ages 30–50, non-missing family income + father's occupation
- Race: White and Black only (other races < 1% pre-1950)
- Weights: Reweighted to census race $\times$ sex shares per birth decade

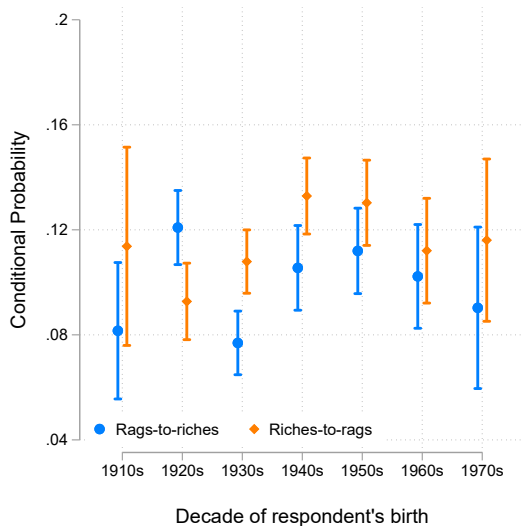
Parental Income Imputation:

- Predict $\hat{Y}^P$ using 28 harmonized occupation categories $\times$ race $\times$ South
- Sources: 1901 Cost of Living Survey, 1936 Expenditure Survey, 1940–90 Censuses
- Matched to respondent's ≈ 10 th birthday; special adjustments for farmers

Sample Sizes by Birth Decade:

1910s	1920s	1930s	1940s	1950s	1960s	1970s
4,641	11,857	11,426	10,288	9,531	5,691	2,668

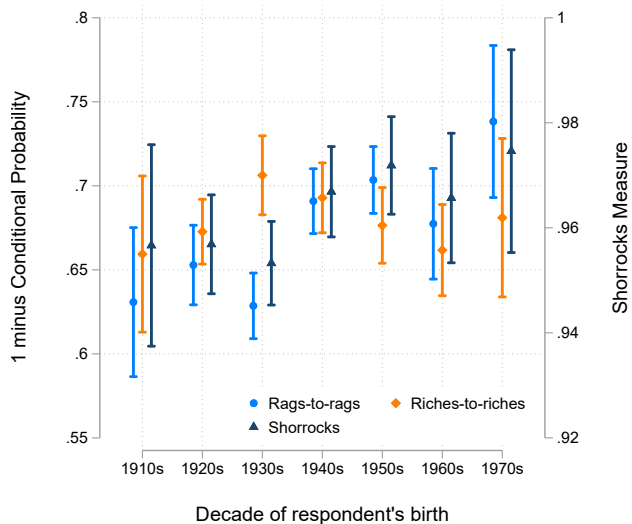
Transition Matrix Measures Co-Move

[◀ Back](#)

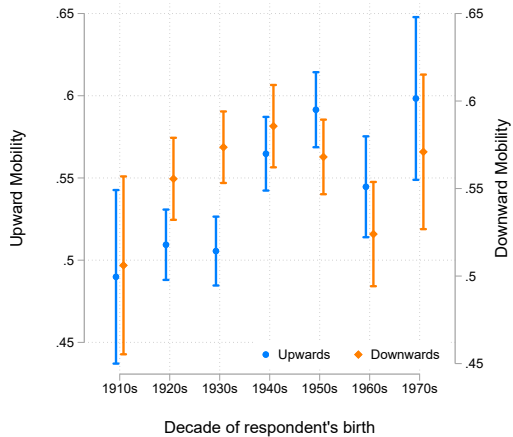
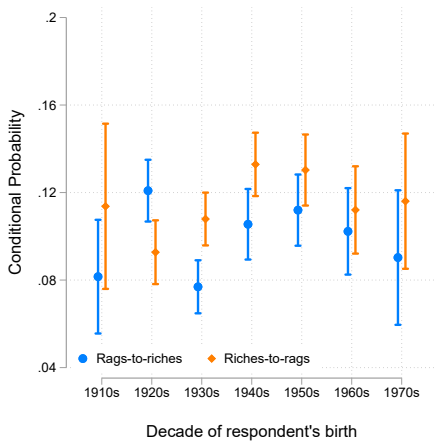
- Quintile Transition Matrix

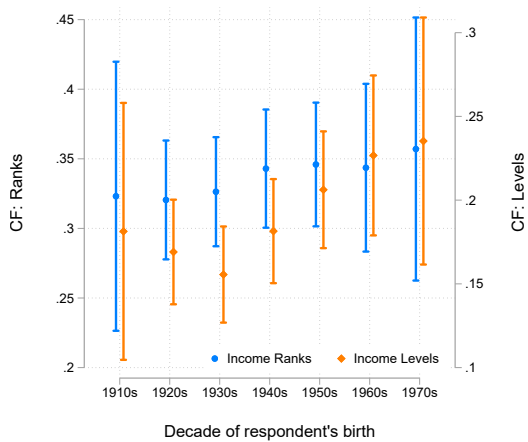
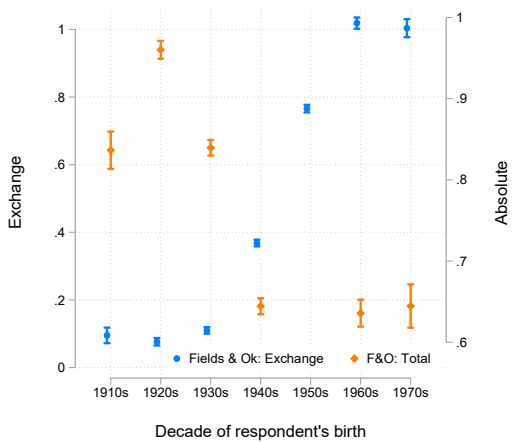
- Q1→Q5 & Q5→Q1

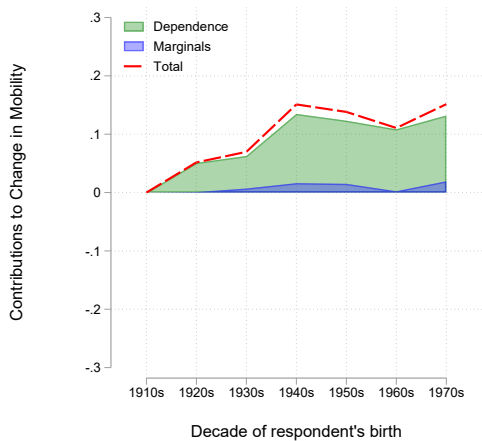
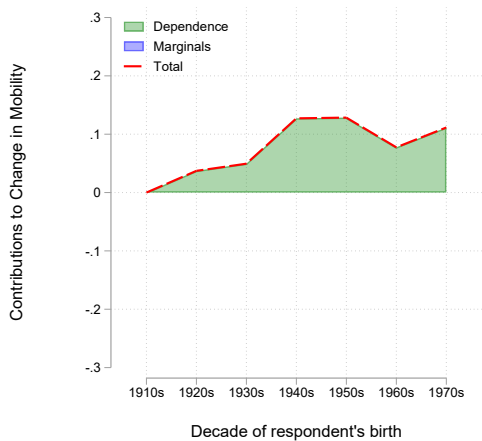
Transition Matrix Measures Co-Move [◀ Back](#)

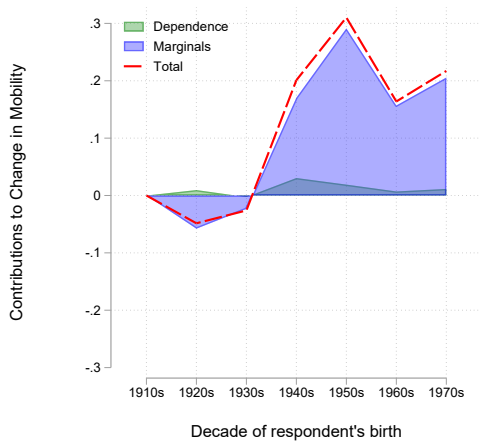
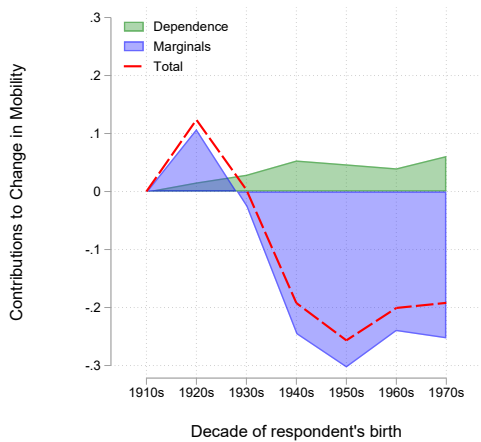


- Quintile Transition Matrix
- Q5→Q5 & Q1→Q1 & Trace









Testing The Concordance Order [◀ Back](#)

Set up as a moment inequality testing procedure (Andrews and Soares, 2010)

- Moments on a grid $\mathcal{G}$:

$$m_g = C_A(u_g, v_g) - C_B(u_g, v_g), \quad g = 1, \dots, G.$$

- Null hypothesis of dominance

$$H_0: m_g \geq 0 \text{ for all } g = 1, \dots, G \quad \text{vs.} \quad H_1: m_g < 0 \text{ for some } g.$$

- Test statistic is the minimum of the studentized moments:

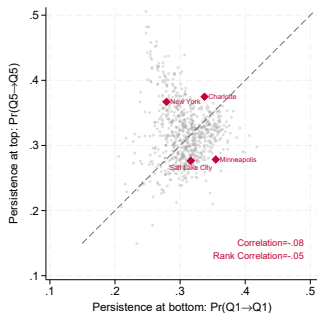
$$T_n = \min_{g=1, \dots, G} \frac{\hat{m}_g}{\hat{s}_g},$$

w/ bootstrap standard error $\hat{s}_g$

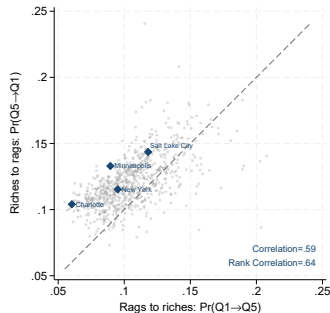
- Test $A \succeq B$ and $B \succeq A$

Spatial Mobility: Measure Disagreement Across CZs [◀ Back](#)

Chetty et al. (2014) CZ transition matrices, remapped to within-CZ copulas



Q1→Q1 vs. Q5→Q5 persistence

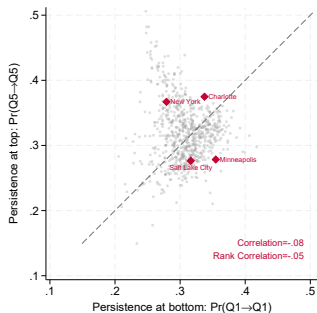


Q1→Q5 vs. Q5→Q1 tail mobility

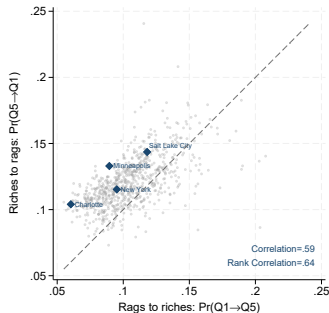
- Salt Lake $\succ$ Charlotte on *every* measure...
 - ...but Salt Lake vs. NYC, NYC vs. Minneapolis: depends on measure

Spatial Mobility: Measure Disagreement Across CZs [◀ Back](#)

Chetty et al. (2014) CZ transition matrices, remapped to within-CZ copulas



Q1→Q1 vs. Q5→Q5 persistence



Q1→Q5 vs. Q5→Q1 tail mobility

- Salt Lake $\succ$ Charlotte on *every* measure...
 - ...but Salt Lake vs. NYC, NYC vs. Minneapolis: depends on measure
- Top-quartile by one persistence measure $\neq$ top-quartile by the other in **91%** of CZs
 - **61%** for the tail-mobility pair

A Simple Illustrative Example [◀ Back](#)

Separating Dependence & Inequality

Assume Incomes jointly Log-Normal (closed form)

$$\begin{pmatrix} \ln Y^K \\ \ln Y^P \end{pmatrix} \sim N \left[\begin{pmatrix} \mu_K \\ \mu_P \end{pmatrix}, \begin{pmatrix} \sigma_K^2 & \rho\sigma_K\sigma_P \\ \rho\sigma_K\sigma_P & \sigma_P^2 \end{pmatrix} \right]$$

A Simple Illustrative Example [◀ Back](#)

Separating Dependence & Inequality

Assume Incomes jointly Log-Normal (closed form)

$$\begin{pmatrix} \ln Y^K \\ \ln Y^P \end{pmatrix} \sim N \left[\begin{pmatrix} \mu_K \\ \mu_P \end{pmatrix}, \begin{pmatrix} \sigma_K^2 & \rho\sigma_K\sigma_P \\ \rho\sigma_K\sigma_P & \sigma_P^2 \end{pmatrix} \right]$$

What do measures depend on?

A Simple Illustrative Example [◀ Back](#)

Separating Dependence & Inequality

Assume Incomes jointly Log-Normal (closed form)

$$\begin{pmatrix} \ln Y^K \\ \ln Y^P \end{pmatrix} \sim N \left[\begin{pmatrix} \mu_K \\ \mu_P \end{pmatrix}, \begin{pmatrix} \sigma_K^2 & \rho\sigma_K\sigma_P \\ \rho\sigma_K\sigma_P & \sigma_P^2 \end{pmatrix} \right]$$

What do measures depend on?

- Rank-rank slope: $\text{Corr}(R^K, R^P) = 6/\pi \arcsin(\rho/2) \approx \rho$ (Kruskal, 1958) *only dependence!*

A Simple Illustrative Example [◀ Back](#)

Separating Dependence & Inequality

Assume Incomes jointly Log-Normal (closed form)

$$\begin{pmatrix} \ln Y^K \\ \ln Y^P \end{pmatrix} \sim N \left[\begin{pmatrix} \mu_K \\ \mu_P \end{pmatrix}, \begin{pmatrix} \sigma_K^2 & \rho\sigma_K\sigma_P \\ \rho\sigma_K\sigma_P & \sigma_P^2 \end{pmatrix} \right]$$

What do measures depend on?

- Rank-rank slope: $\text{Corr}(R^K, R^P) = 6/\pi \arcsin(\rho/2) \approx \rho$ (Kruskal, 1958) *only dependence!*
- IGE: $\beta = \rho \cdot \frac{\sigma_K}{\sigma_P}$ *dependence and inequality*

$$\begin{aligned} \text{- Changes: } \Delta\beta = & \underbrace{(\rho^B - \rho^A) \cdot \frac{\sigma_K^B}{\sigma_P^B}}_{\text{change in rank correlation}} + \underbrace{\rho^A \cdot \left(\frac{\sigma_K^B}{\sigma_P^B} - \frac{\sigma_K^A}{\sigma_P^A} \right)}_{\text{change in inequality}} \end{aligned}$$

A Simple Illustrative Example [◀ Back](#)

Separating Dependence & Inequality

Assume Incomes jointly Log-Normal (closed form)

$$\begin{pmatrix} \ln Y^K \\ \ln Y^P \end{pmatrix} \sim N \left[\begin{pmatrix} \mu_K \\ \mu_P \end{pmatrix}, \begin{pmatrix} \sigma_K^2 & \rho \sigma_K \sigma_P \\ \rho \sigma_K \sigma_P & \sigma_P^2 \end{pmatrix} \right]$$

What do measures depend on?

- Rank-rank slope: $\text{Corr}(R^K, R^P) = 6/\pi \arcsin(\rho/2) \approx \rho$ (Kruskal, 1958) **only dependence!**

- IGE: $\beta = \rho \cdot \frac{\sigma_K}{\sigma_P}$ **dependence and inequality**

$$\text{Changes: } \Delta\beta = \underbrace{(\rho^B - \rho^A) \cdot \frac{\sigma_K^B}{\sigma_P^B}}_{\text{change in rank correlation}} + \underbrace{\rho^A \cdot \left(\frac{\sigma_K^B}{\sigma_P^B} - \frac{\sigma_K^A}{\sigma_P^A} \right)}_{\text{change in inequality}}$$

- Rising inequality ($\sigma_K/\sigma_P \uparrow$) can **dominate** IGE even as rank dependence **weakens** ($\rho \downarrow$)